

EPITALON

Synthetic Tetrapeptide | Ala-Glu-Asp-Gly | MW: ~390.35 Da

WHAT IS EPITALON?

A synthetic analog of epithalamin, a natural polypeptide extracted from the pineal gland. Designed to replicate the active fraction responsible for pineal gland signaling. Primarily studied for anti-aging and telomerase activation.

MECHANISM OF ACTION

Telomerase Activation

Upregulates TERT, elongating telomeres in somatic cells, delaying replicative senescence.

Antioxidant Defense

Enhances SOD and catalase expression, reducing ROS.

Neuroendocrine Regulation

Influences pineal gland and HPA axis, improving melatonin secretion and circadian rhythm regulation.

Gene Expression Modulation

Alters genes involved in stress response, apoptosis, and cell cycle regulation.

CONDITIONS STUDIED

- Aging and longevity (rodent life-extension models)
- Telomere shortening disorders
- Neurodegeneration (Alzheimer's, Parkinson's models)
- Cancer models (anti-tumor via cell cycle normalization)
- Circadian dysfunction and sleep disorders
- Immune dysfunction in aged populations

SAFETY PROFILE

No acute toxicity reported in animal studies, even with prolonged administration. No LD50 reported within tested ranges. Theoretical concern: telomerase activation could theoretically promote oncogenesis, though anti-tumor effects have also been observed. Long-term safety undefined.

BIOLOGICAL ACTIVITY

- Stimulates telomerase activity in somatic cells
- Regulates neuroendocrine function (pineal gland)
- Reduces oxidative stress and lipid peroxidation
- Modulates circadian rhythm and melatonin secretion
- Potential lifespan extension in animal models

MAIN PURPOSE

Designed to mimic pineal gland peptides that regulate aging, circadian biology, and DNA protection. Intended to slow aging processes, improve cellular longevity, and support endocrine and immune function.

DRUG INTERACTIONS

- No major CYP450 interactions expected
- Theoretical synergism with melatonin or other circadian-regulating agents
- Caution with immunosuppressants (may counteract immune-modulating effects)

CONTRAINDICATIONS & PRECAUTIONS

- Known or suspected malignancy (telomerase risk)
- Pregnancy and lactation (not studied)
- Use with caution in autoimmune conditions
- Not FDA/EMA approved; research peptide only

REGULATORY STATUS

Not approved by FDA or EMA for medical use.

Used experimentally in Russia for gerontology research.

Not WADA-listed but could fall under unapproved substances. Classified as a research peptide globally.

KEY RESEARCH

- Araj et al. (2025) PMC overview of Epitalon
- Liu et al. (2022) oocyte aging protection
- Khavinson & Anisimov (2020) gene expression and neuroprotection (PMC7037223)

Want to explore if peptide therapy fits your health goals? DM @mugdhapradhan_official

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